

The Human Body



The Nervous System

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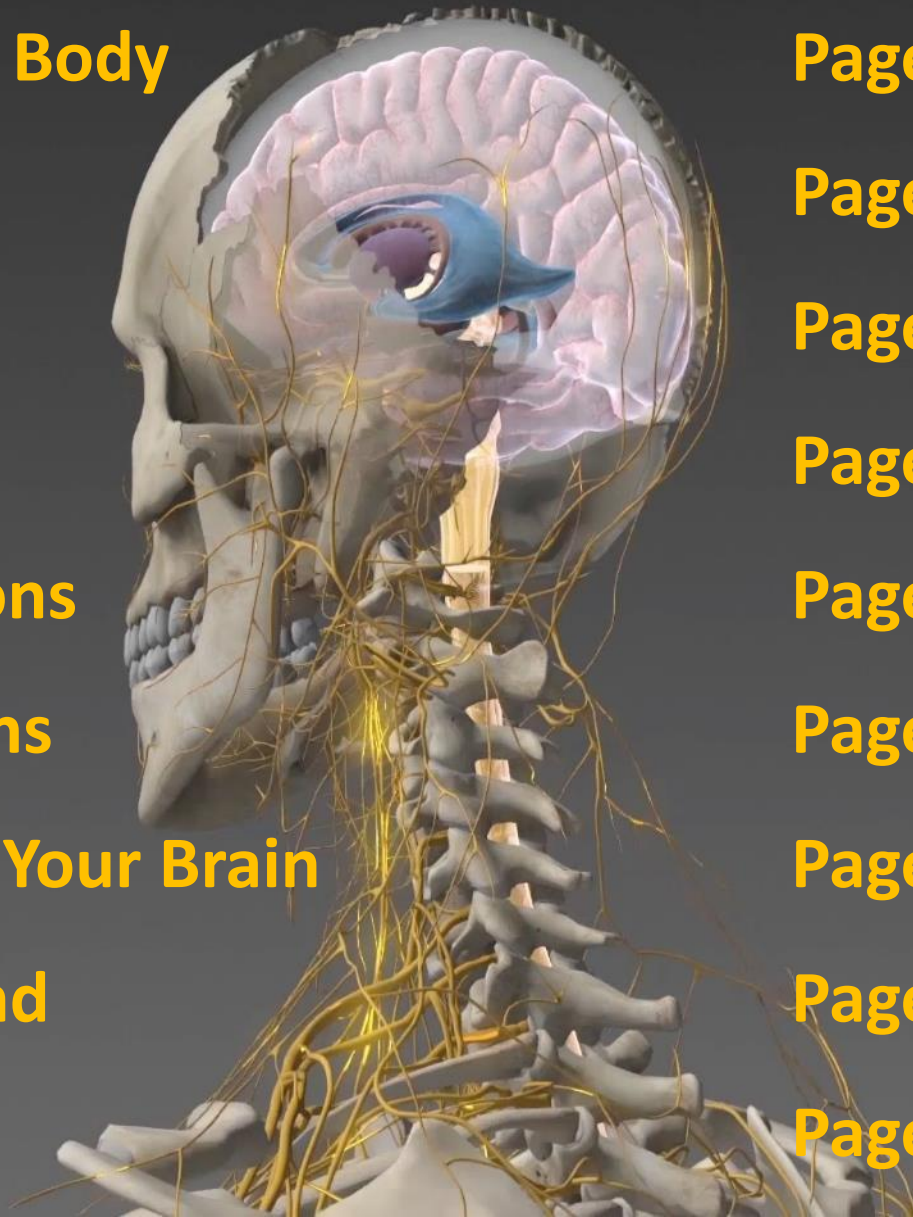
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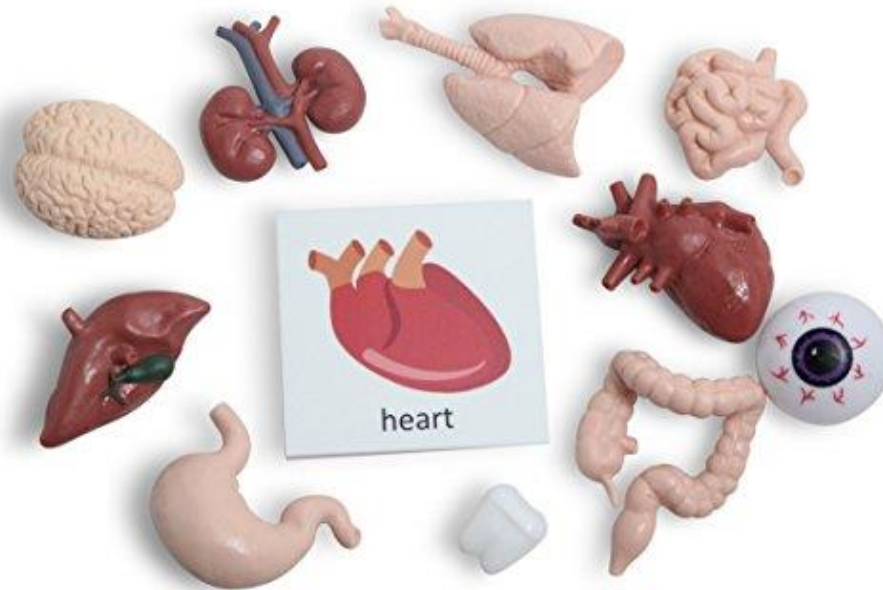
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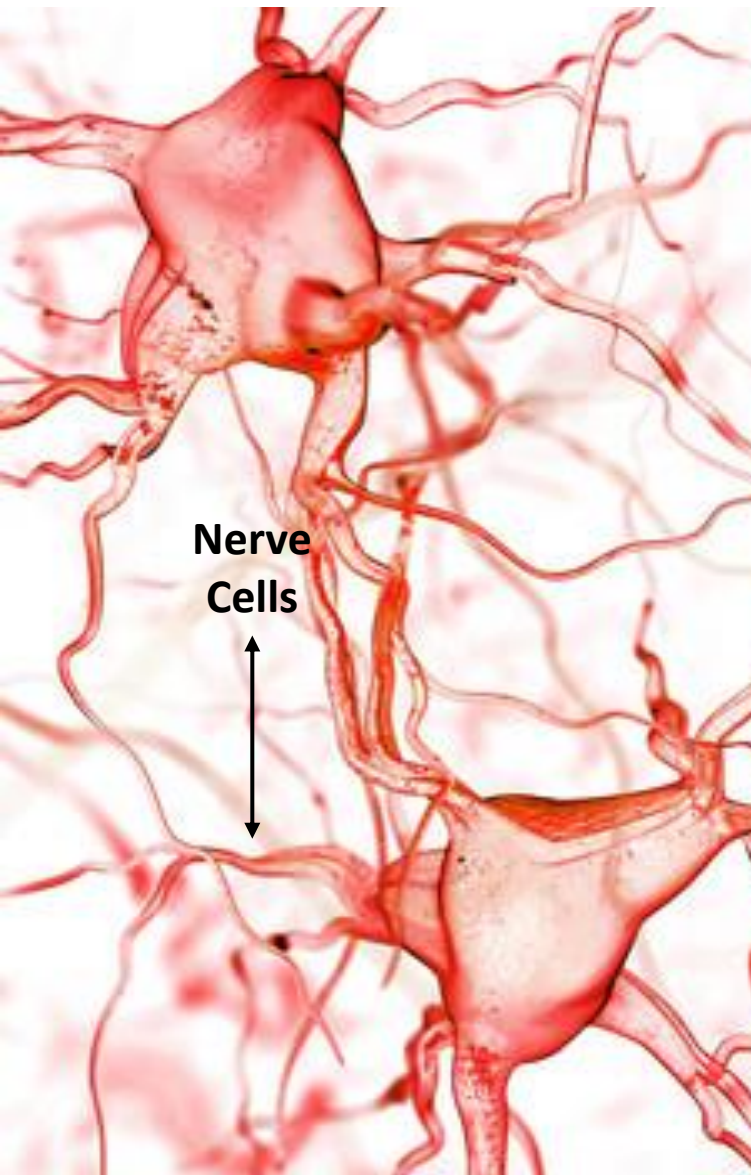


THE HUMAN BODY

The human body is a wonderfully efficient and complex machine. It has organs and systems that are organized to perform specific functions. The stomach, the intestines, the liver and the pancreas help in the digestion of food and constitute the digestive system. The lungs help in breathing and make up the respiratory system. The system that helps you feel and react is the nervous system. The system that helps in movement is the locomotory/muscular system.



NERVE CELLS



Have you ever wondered why you never feel any sensation when you cut your nails or hair but feel pain when you get a cut on your finger?

When a message reaches the brain from any part of the body, the brain interprets it and tells the body how to react.

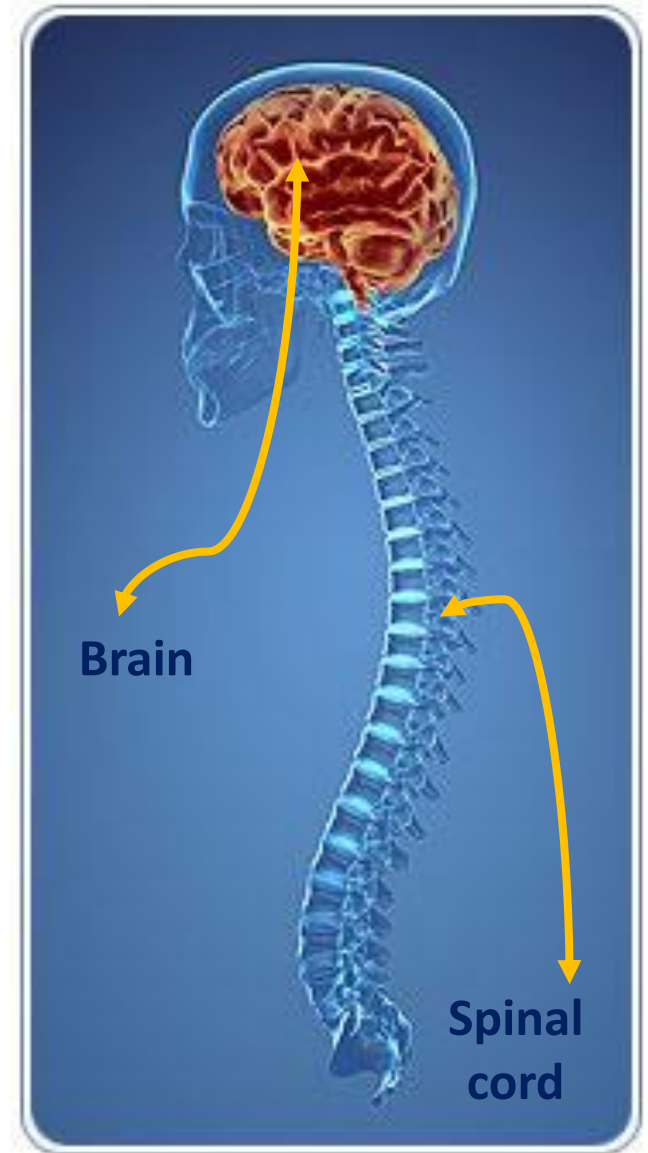
For example, if you accidentally cut your finger on a sharp object, special cells called **nerve cells** that are present in your skin, shoot a message of pain to your brain. The brain tells you to pull your hand away. This happens because of the functioning nervous system in your body.

NERVE CELLS

The brain, the spinal cord and the nerves form the nervous system.

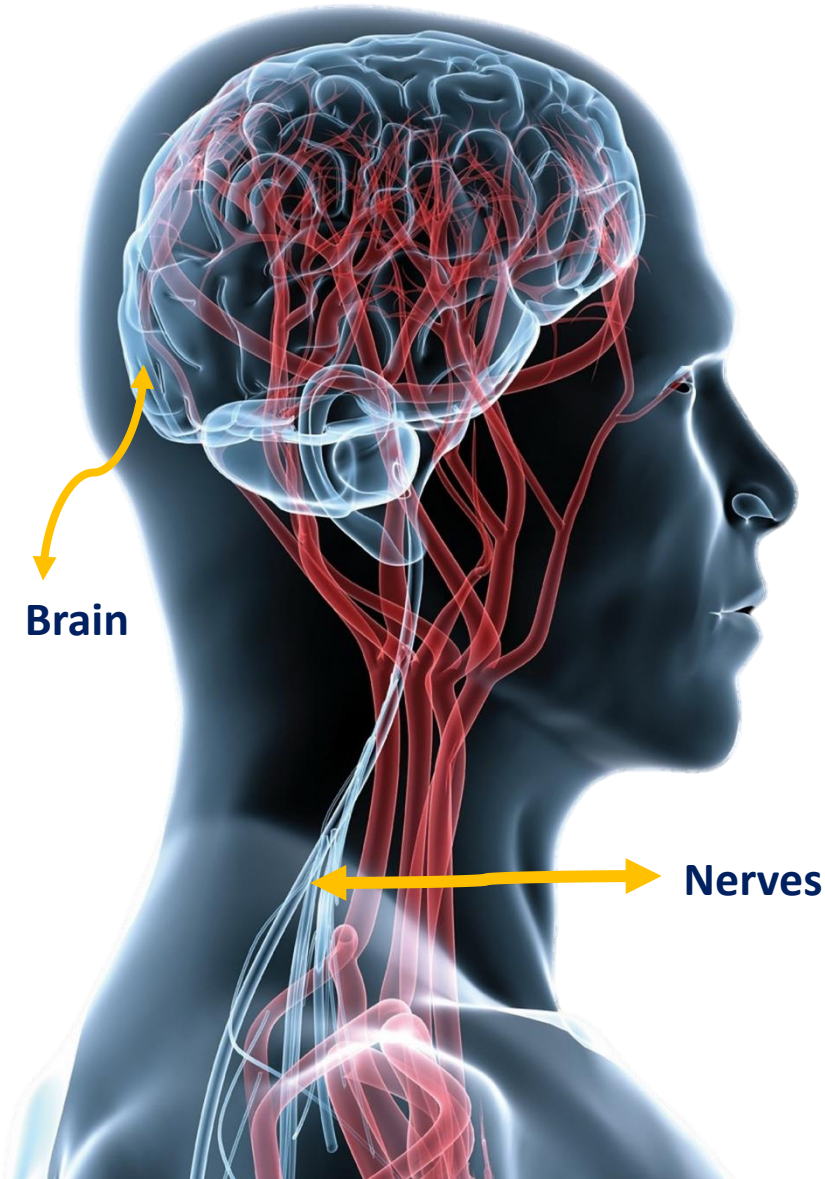
Nerves are like telephone wires that link every part of the cells or **neurons** which are connected to each other.

Neurons receive messages from the outside environment and pass them on to the brain through the network of nerves in the body.



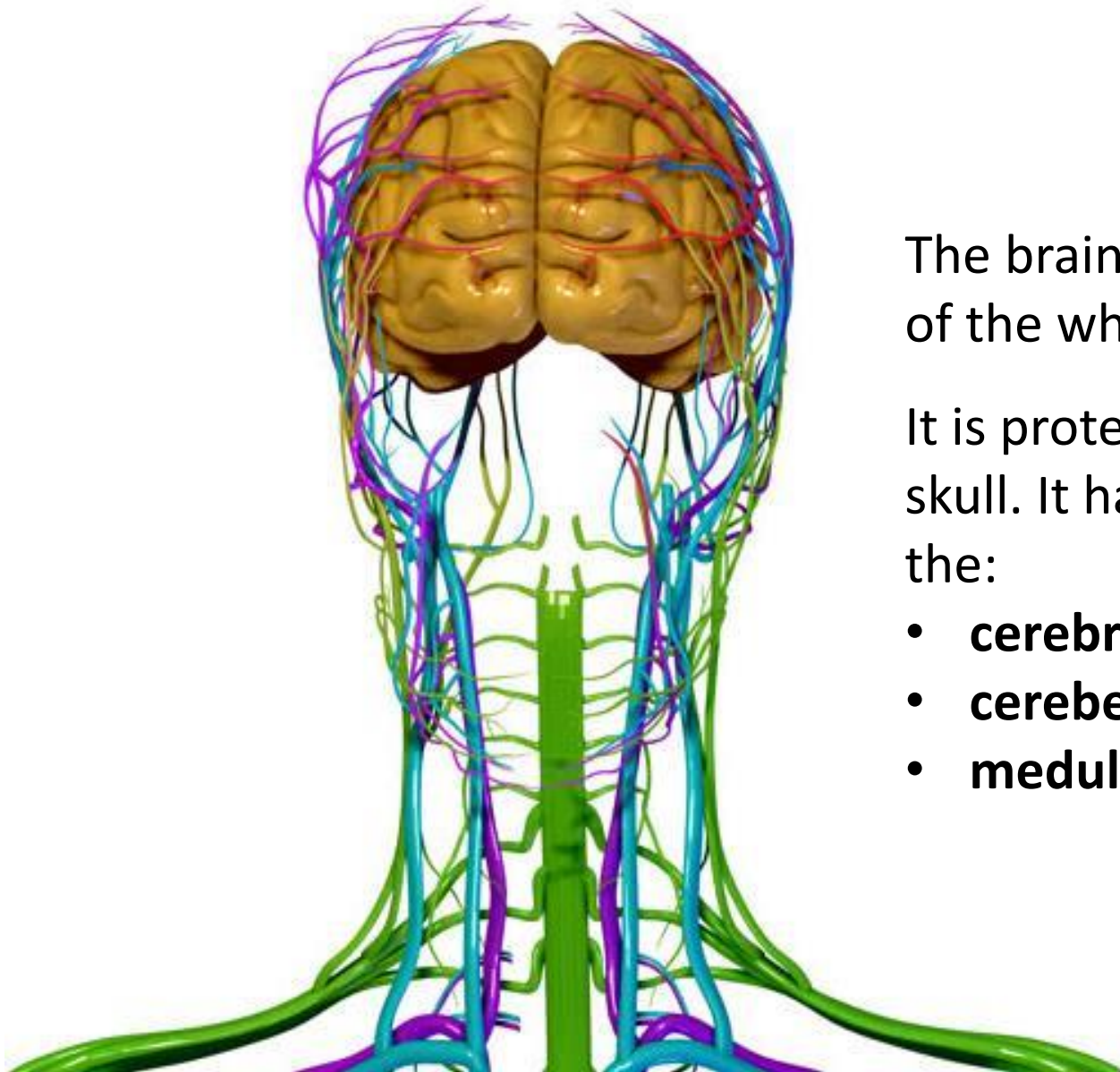
Wizard's Corner

Messages travel along the large nerves, at a speed of more than 90 metres per second.



You know that your hair and nails, grow gradually. As they grow, their upper part becomes dead. So, there are no neurons present in them. This is why we do not feel any sensation while cutting them.

The **brain** is like a telephone exchange where messages are sent and received through the nerves. The **spinal cord** can be compared to a bundle of telephone lines inside a telephone cable.



The brain is the control centre of the whole body.

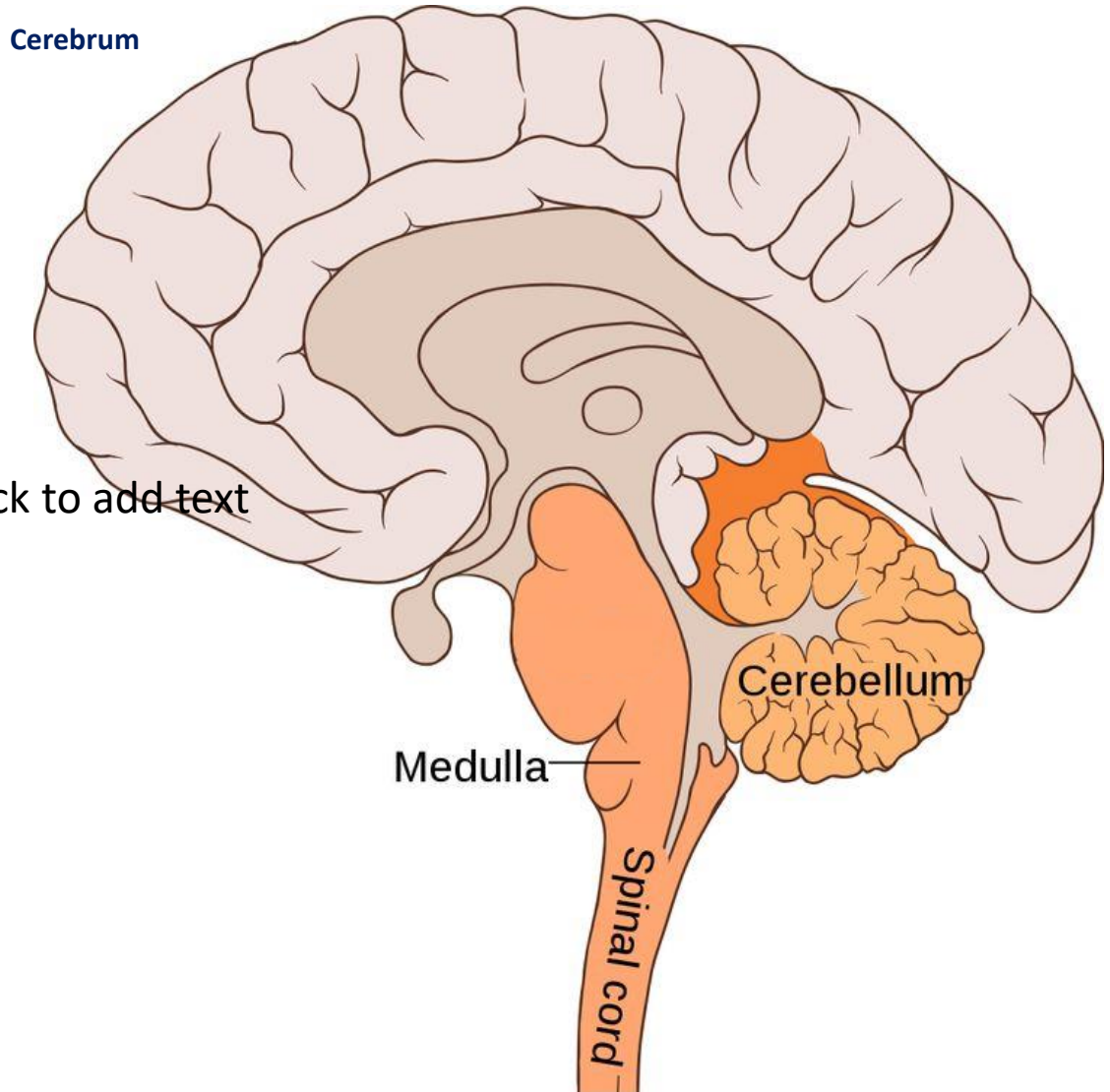
It is protected by a hard, bony skull. It has three main parts, the:

- **cerebrum**
- **cerebellum**
- **medulla**

THE CEREBRUM

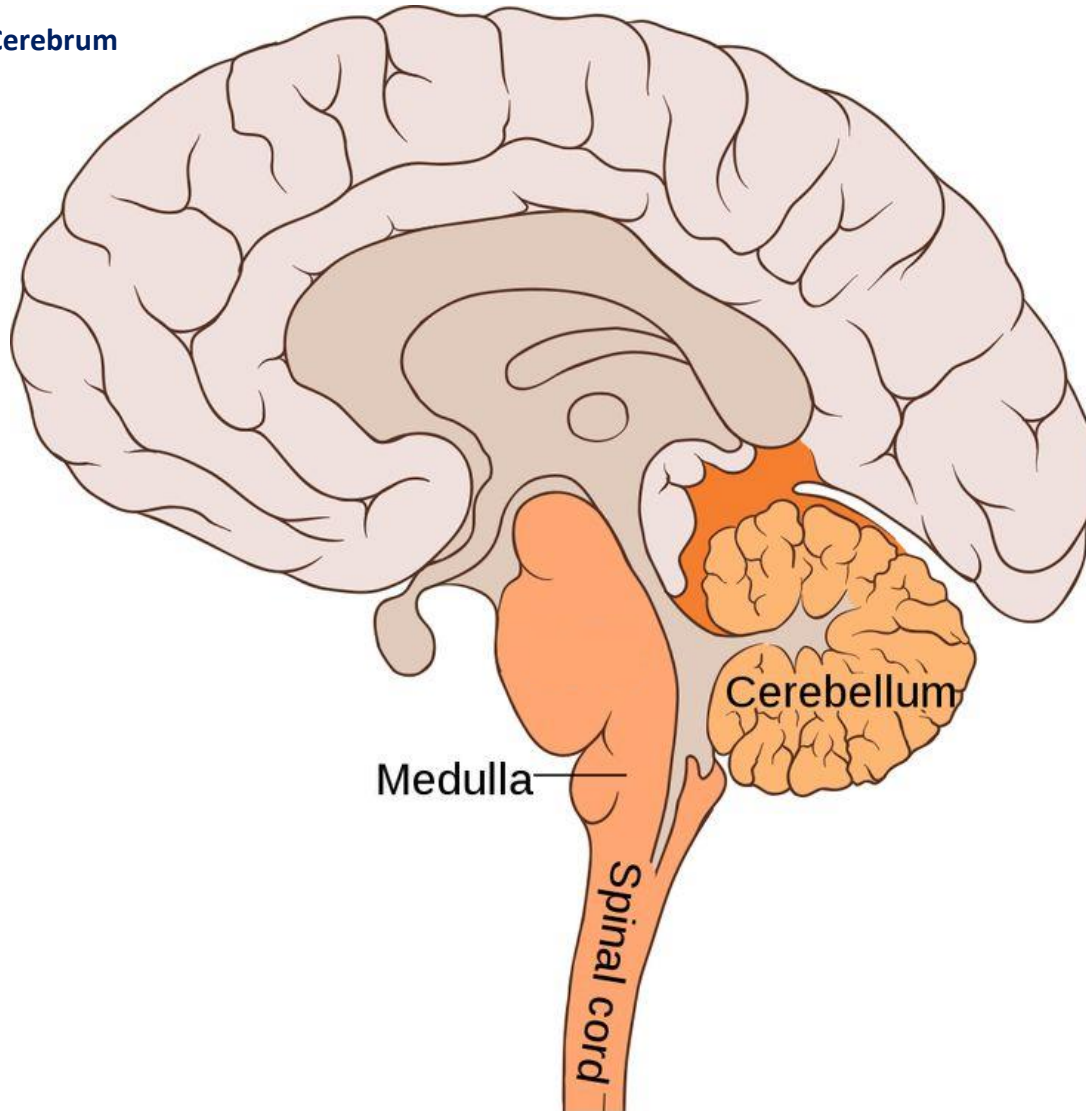
It is the largest part of the brain. It has many folds and grooves. It receives messages from the sense organs and decides what the body should do.

It is responsible for our memory, intelligence, logic and learning.



THE CEREBELLUM

Cerebrum



It is a small part of the brain located below the cerebrum.

It is responsible for the co-ordination of the actions of the muscles. The cerebellum helps in maintaining the balance of the body. It helps us to run, walk, play, etc.

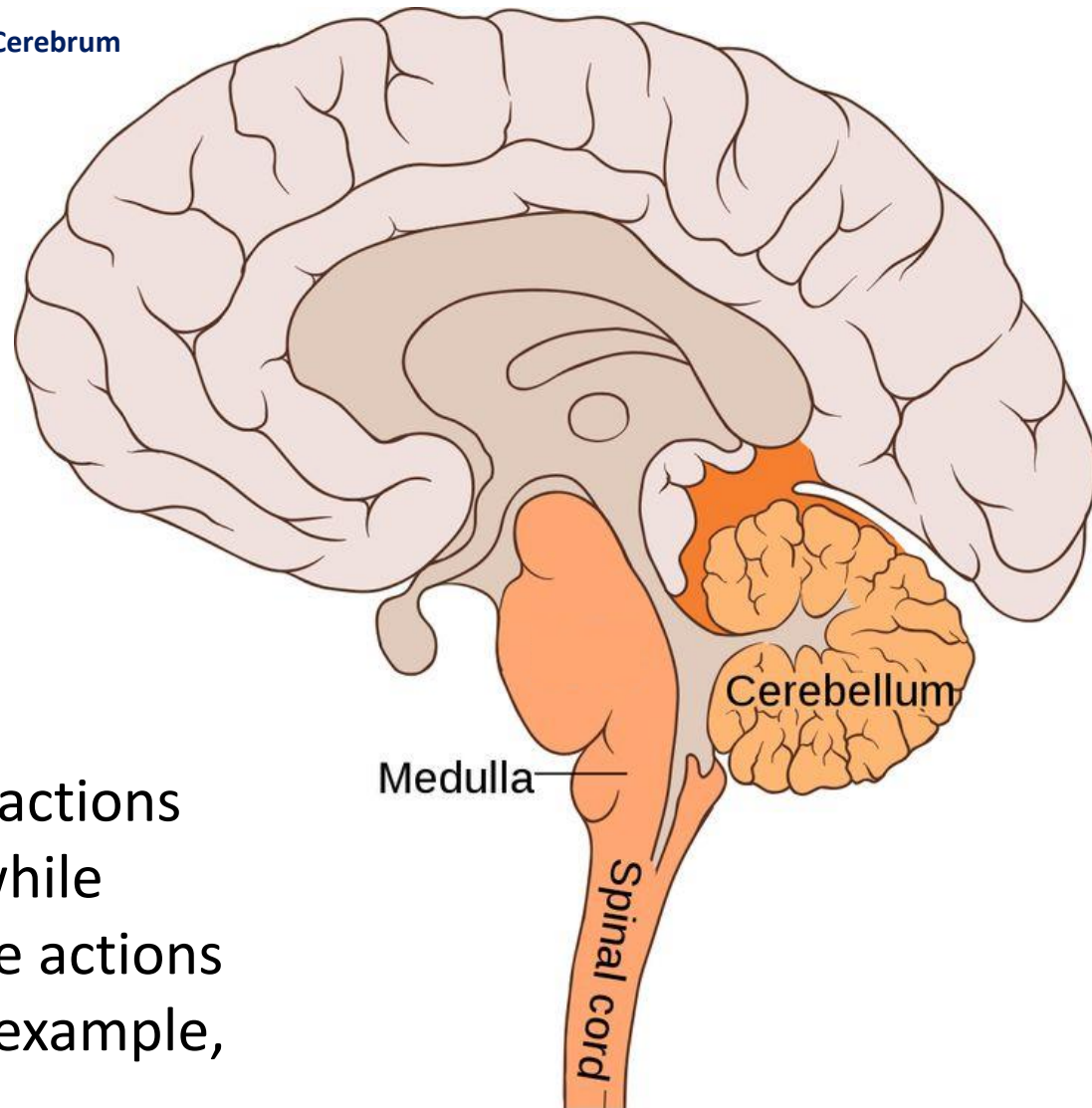
THE MEDULLA

It is also called the **brain stem**. It is situated at the base of the brain and joins the spinal cord.

It controls involuntary actions of the body such as breathing, the heartbeat and the working of many internal organs.

Voluntary actions are those actions that are under our control while involuntary actions are those actions that we cannot control. For example, the beating of our heart.

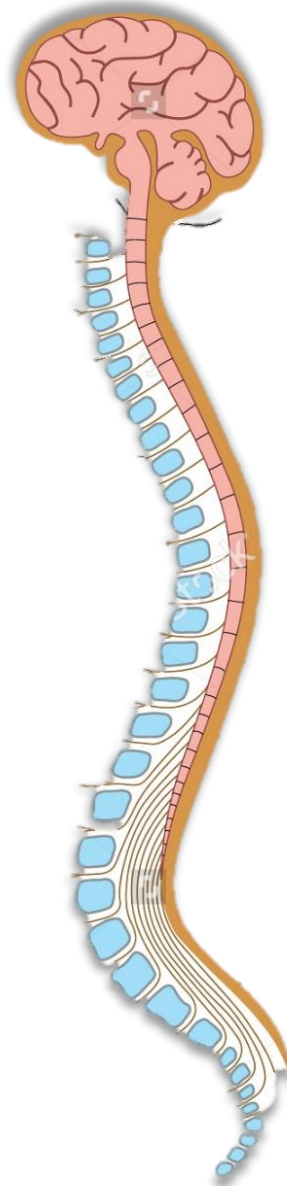
Cerebrum



THE SPINAL CORD

The spinal cord is a thick bundle of nerves.

It extends from the medulla of the brain to almost the end of the spine. Several nerves branch out from the spinal cord to all parts of the body. The spinal cord links the brain with the rest of the body and also controls reflex actions.



Wizard's Corner

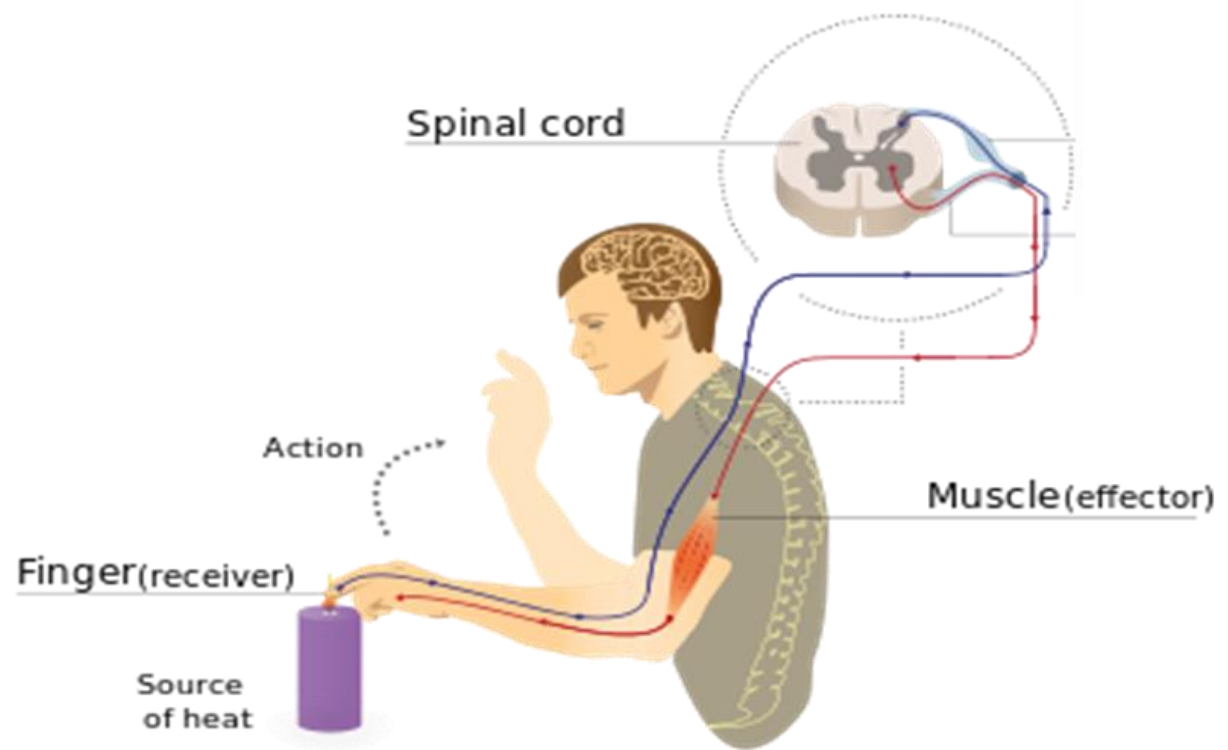
*The brain can store about
100 million bits of
information in a lifetime.*

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NERVES

Messages travel from the sense organs to the brain and from the brain to the spinal cord, to all parts of the body, through the nerves.

For example, when the telephone bell rings, the ear, our sense organ - receives the message in the form of sound waves.



This message is then transmitted to the brain through nerves. The brain interprets the message and sends the order. A message is sent back through the nerves to the muscles of the hand that was used to lift the receiver.



A network of nerves runs throughout the body. Nerves link every part of the body to the brain through the spinal cord. Our body has three kinds of nerves – sensory, motor and mixed.

- 1 **Sensory nerves** carry messages from the sense organs to the spinal cord, or the brain.
- 2 **Motor nerves** carry messages from the brain or the spinal cord to the muscles.
- 3 **Connecting or mixed nerves** carry messages to the brain or spinal cord and bring orders from the brain.

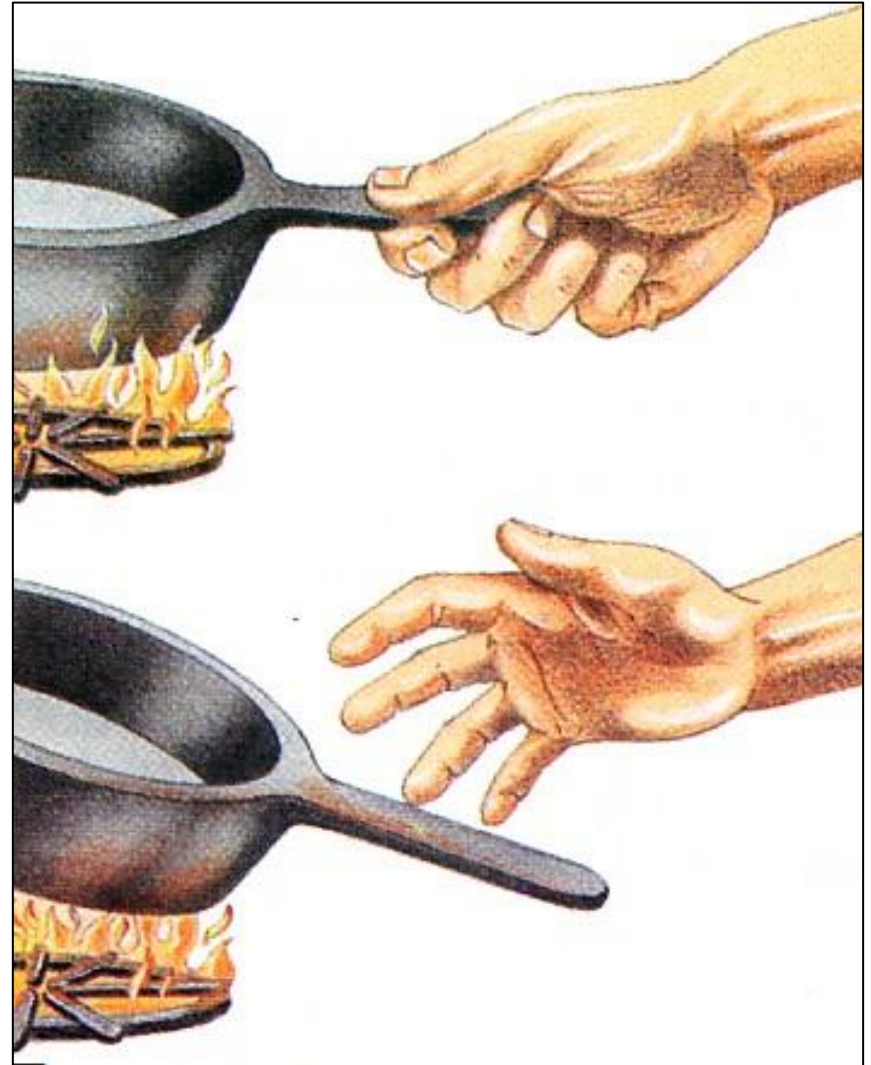


Neurons

REFLEX ACTION

The nervous system helps in controlling all our actions. We cannot read, write, learn, dance, play, eat and go to school without the functioning of the nervous system of our body.

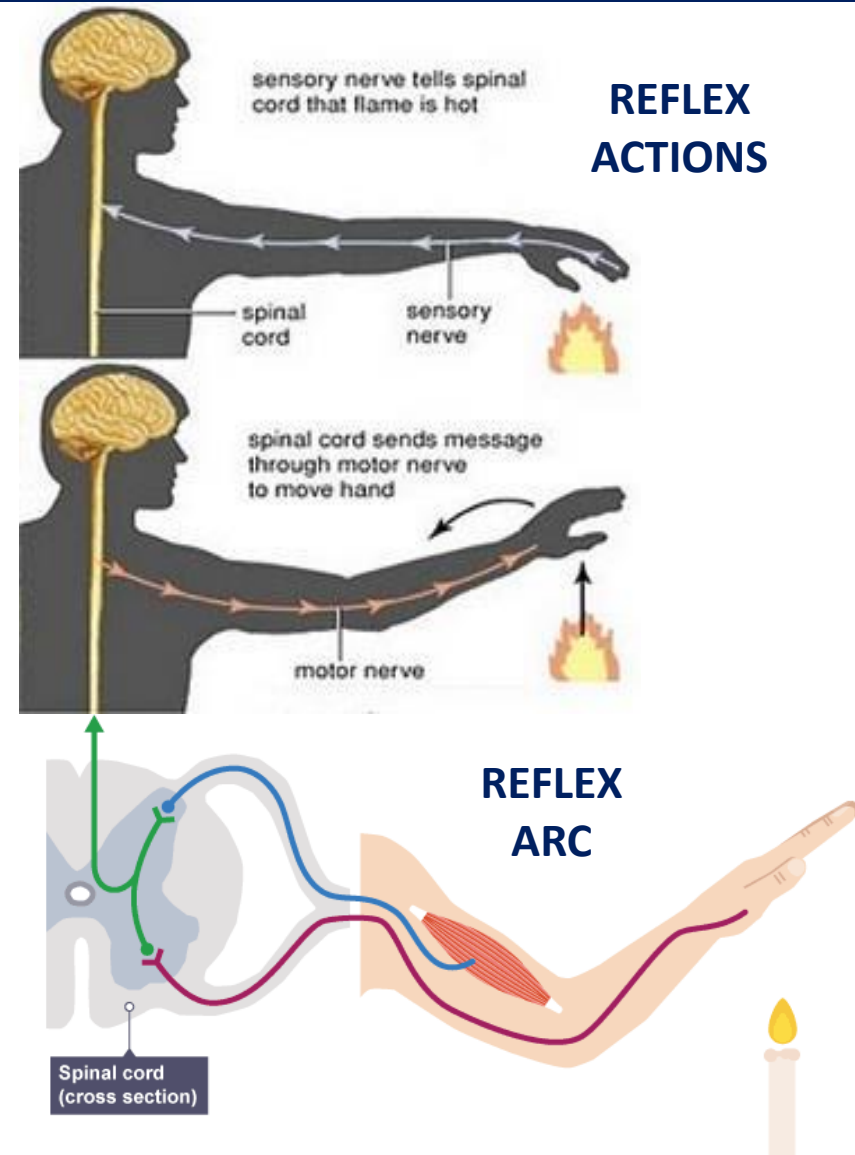
Apart from the regular actions, some of our sudden actions are also controlled by our nervous system. Such actions are known as **reflex actions**.



REFLEX ACTION

Reflex actions are sudden, involuntary actions in response to a stimulus. A **stimulus** is any change in the external or internal environment of an organism.

For example, if you touch a hot object, you withdraw your hands quickly or, if you see your favourite food, your mouth waters. Messages to carry out such actions do not reach the brain as they are transmitted through the spinal cord. So, we do not need time to think about these actions as they happen instantly.



SENSE ORGANS



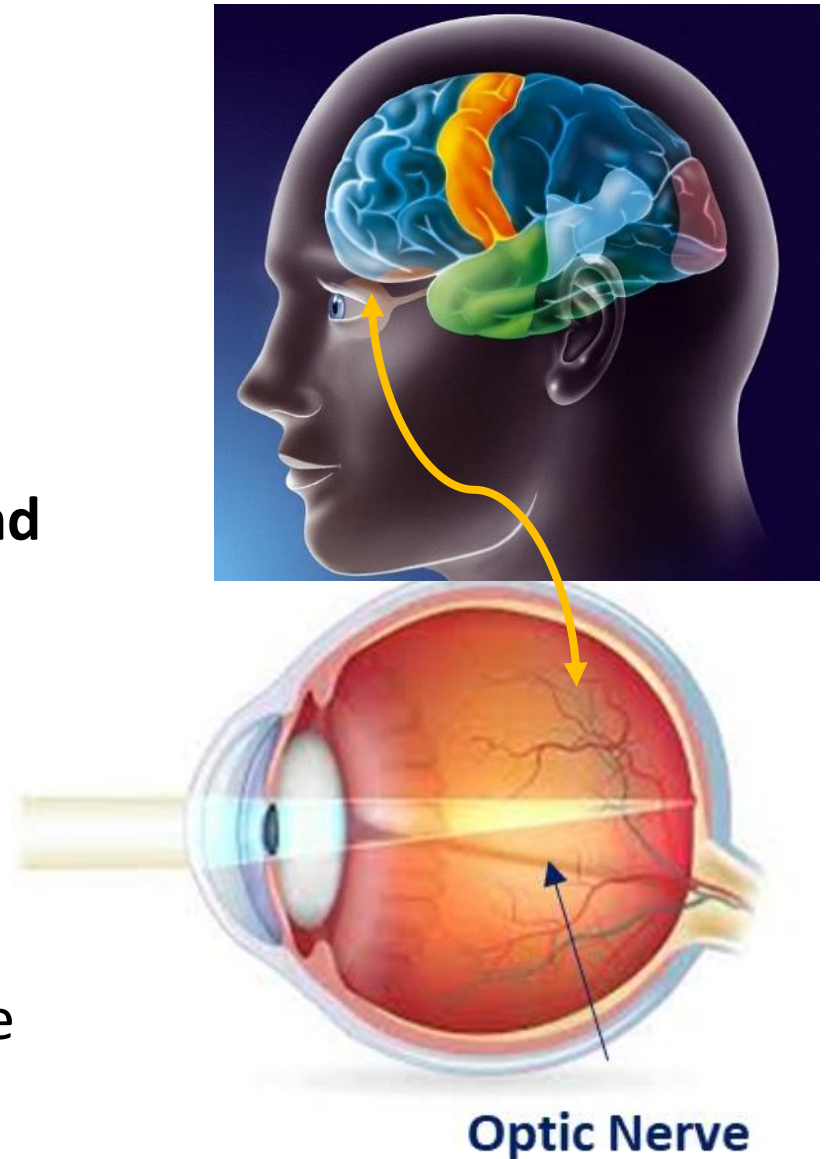
We become aware of our surroundings by seeing, hearing, feeling, etc. We use our senses to obtain information about our environment.

The sense organs are, thus, our windows to our world. These include the eyes, the ears, the nose, the skin and the tongue.



THE EYES

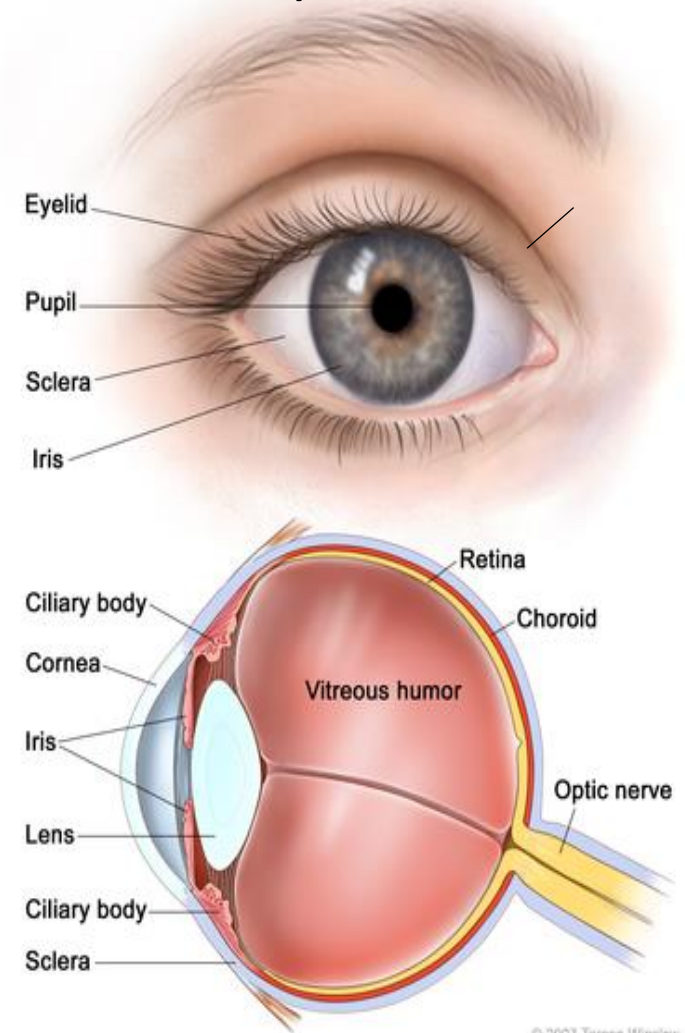
- 1 We get most of the information about the environment through our eyes. Eyes are considered as the most important sense organ.
- 2 The eyes are set into the eye socket in the skull. Our **eyelids and eyelashes** protect our eyes from dirt, strong light, and injuries.
- 3 The eyes are connected to the brain by the optic nerve that carries the message to the brain, and the brain tells us what we are looking at.



PARTS OF AN EYE

Eyeballs, which rest on a pad of fat have three main layers.

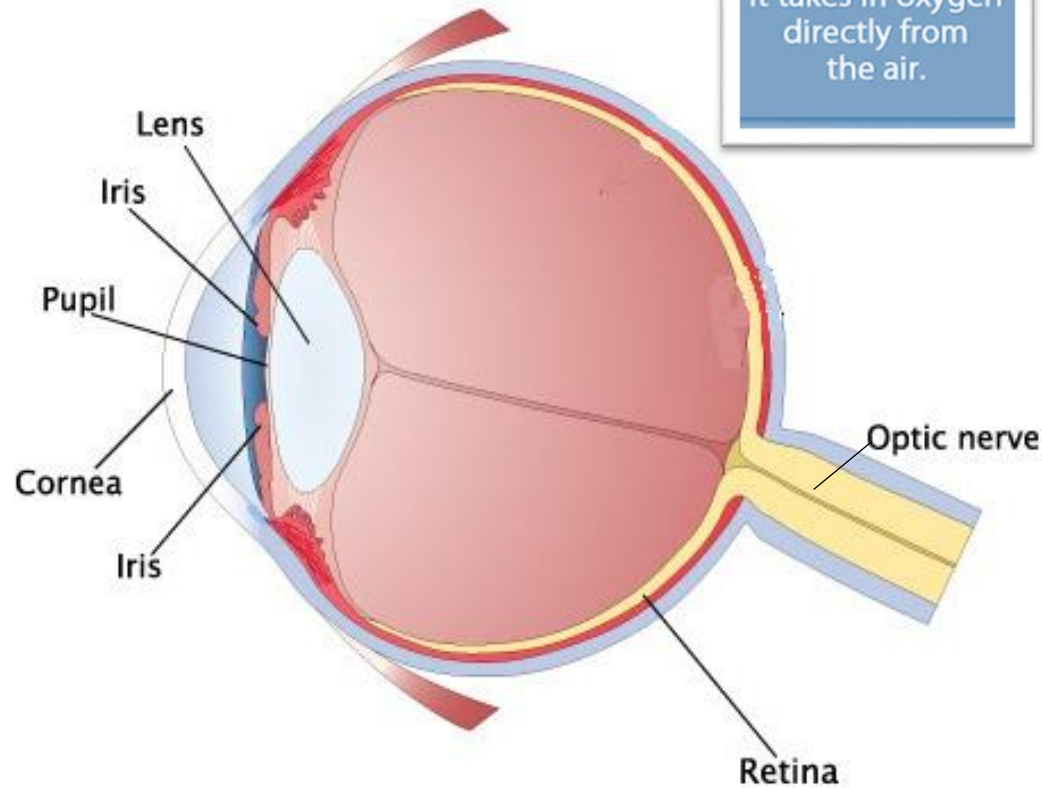
- 1 The **cornea**, the first layer, is transparent.
- 2 The second layer is coloured and is called the **iris**. The black spot in the centre of the iris is the **pupil**. Tiny muscles control the size of the pupil depending on the amount of light the eyes are exposed to. If there is less light, the pupil automatically becomes bigger to take in more light; and if the light is bright, the pupil becomes smaller to take in less light.
- 3 The third layer is the **retina**. It acts as a 'screen' on which the image of an object is formed for us to see it.



HOW DO EYES WORK?

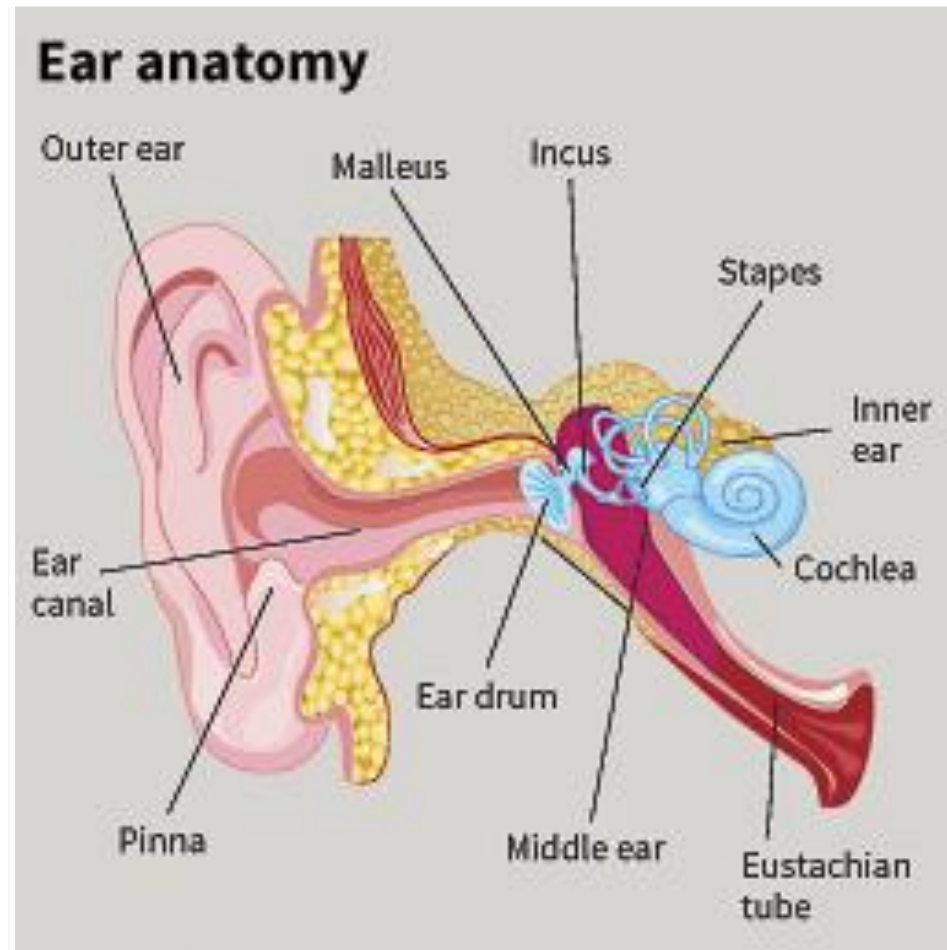
The light coming from an object passes through the cornea and goes to the pupil. Behind the pupil is a lens that helps in focusing the light.

When the light rays fall on the retina, the nerve cells in the retina send a signal to the brain through the **optic nerve**. The image formed on the retina is inverted and the brain processes it in such a way that you perceive the image as right side up.



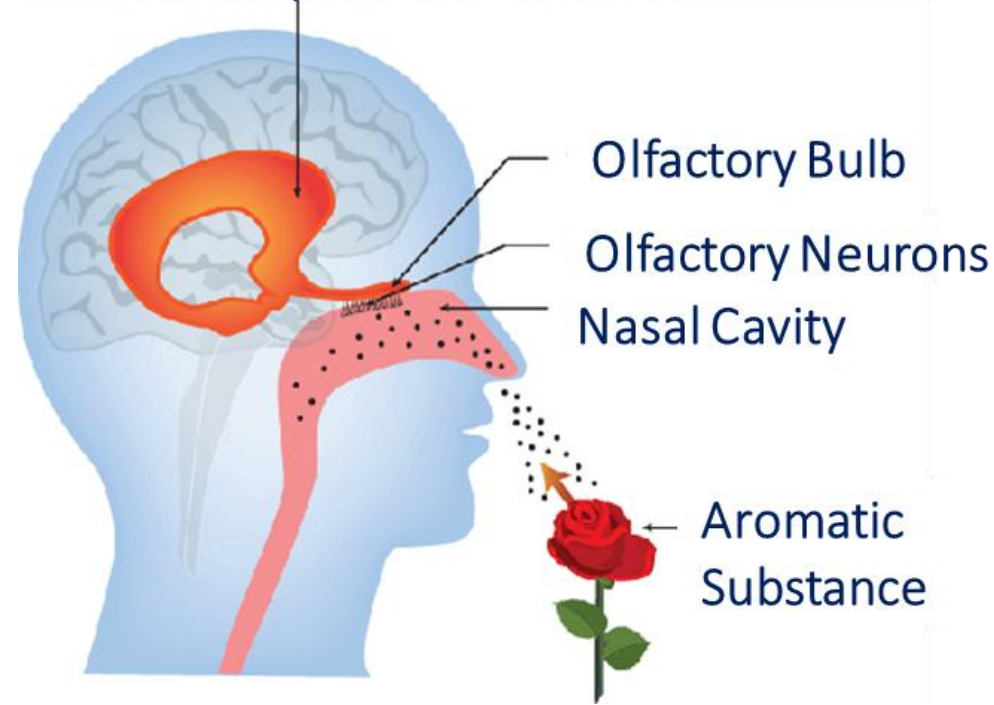
The ears have two functions – hearing and maintaining the balance of the body. The external ear acts like a funnel and collects the sound waves (sound travels in the form of waves).

The sound waves go to the middle part of the ear, which has the ear drum, and three small bones that help in the transmission of sound. The sound waves are then sent to the inner ear, where they are converted into electrical signals by the nerves and sent to the brain, which interprets the message received.



THE NOSE

Limbic System in the Brain

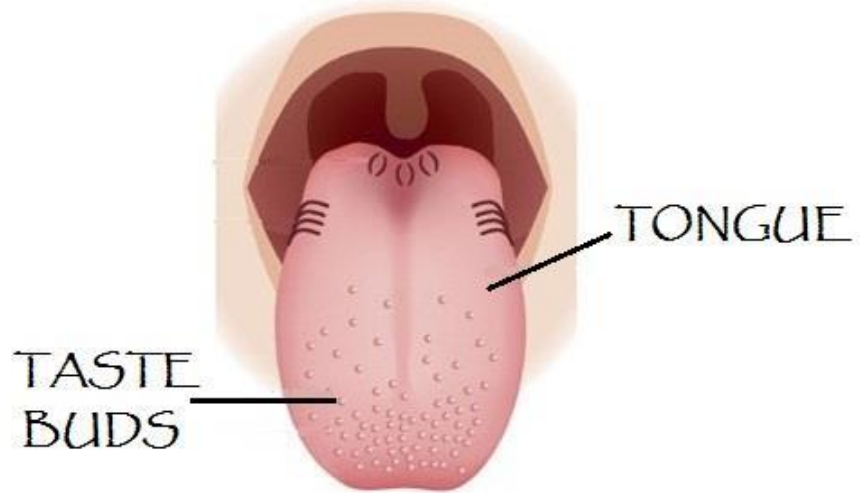


The nose helps us to smell. You know that every time we breathe in, air flows into our nose. The neurons present in the nose are sensitive to smell.

We can sense the smell of different things because of the particles that they are made up of.

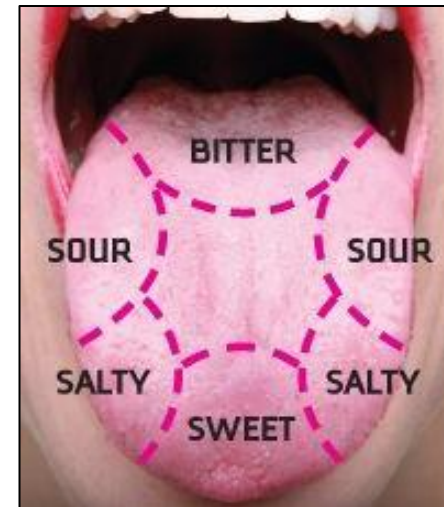
When these particles meet the nerve cells in the nose, they send the message to the brain. It then interprets the message as the smell of the substance. The sense of smell and taste are linked. When food smells good, we want to eat it.

THE TONGUE



The tongue helps us to taste. The tongue has thousands of taste buds which have nerve endings that carry messages to the brain.

We can sense five types of taste – sweet, sour, salty, bitter and umami. The taste buds are grouped together, in different areas of the tongue.



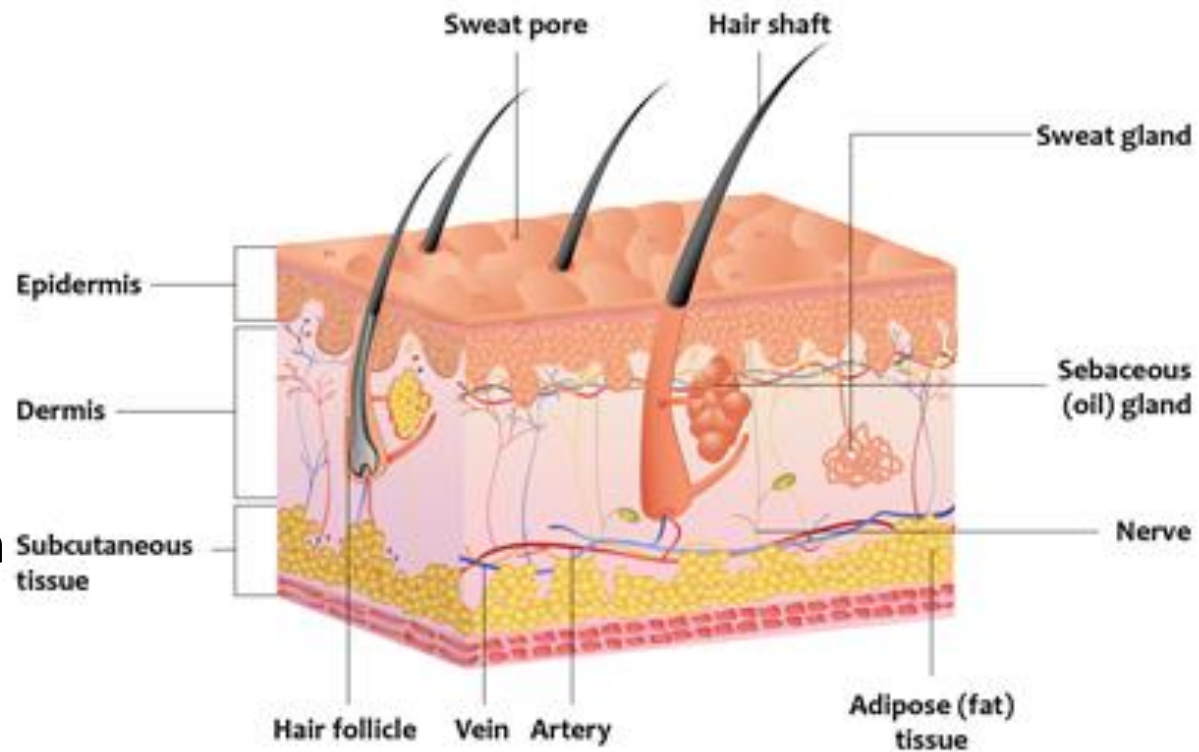
Wizard's Corner

The snake uses its tongue to smell things that is why, it keeps sticking out its tongue.

THE STRUCTURE OF THE SKIN

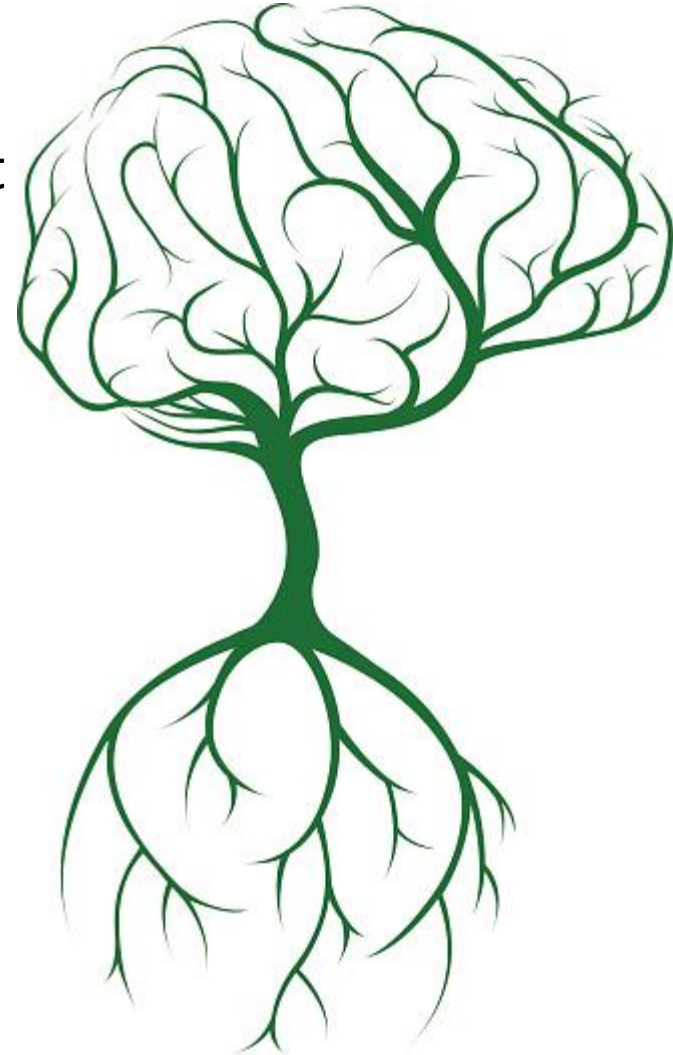
The skin covers the entire body. It helps protect the internal organs of the body. It has millions of nerve endings that can detect sensations such as pain, pressure, heat and cold. The nerves carry these messages to the brain.

The skin also has millions of tiny holes or sweat pores, which allow sweat to come out. Sweat contains body wastes. Thus, the skin helps in cleaning the body. The skin also contains oil glands that release oil. This keeps our hair flexible and makes the skin waterproof.



So, what can you do for your brain?

- 1 Eat healthy foods which contain potassium and calcium, two minerals that are important for the nervous system.
- 2 Get a lot of playtime (exercise).
- 3 Wear a helmet when you ride your bike or play other sports that require head protection.
- 4 Use your brain by doing challenging activities, such as puzzles, reading, playing music, creating art, or anything else that gives your brain a workout!



- 1 The brain, spinal cord and the nerves form the nervous system.
- 2 The brain has three parts: the cerebrum, the cerebellum and the medulla. Each has a specific function.
- 3 There are three kinds of nerves – sensory, motor and mixed.
- 4 The spinal cord runs down from the medulla to the lower back.
- 5 A reflex action is a sudden action which is under the control of the spinal cord.
- 6 We get information about our surroundings through the sense organs which are, eyes, ears, nose, skin and tongue.

Brain: control centre of the body

Nerves: structures that carry messages from one part of the body to another

Reflex action: an involuntary action in response to a stimulus

Taste buds: structures on the surface of the tongue that help us to detect various tastes